ModelIt!

Teacher's Guide

Blood Moon: When the Sky Turns Red

The Cosmic Dance of Earth, Moon, and Sun

Quick Reference

Grade Level	Grade 7
NGSS Standard	MS-ESS1-1, MS-ESS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: MS-ESS1-1, MS-ESS1-2

Develop and use a model of the Earth-Sun-Moon system to describe the cyclic patterns of lunar phases, eclipses of the Sun and Moon, and seasons. Develop and use a model to describe the role of gravity in the motions within galaxies and the solar system.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model of the Earth-Sun-Moon system to explain lunar phases and eclipses.
Disciplinary Core Idea	**ESS1.A/B The Universe and the Solar System** — Patterns of the apparent motion of the Sun, Moon, and stars can be observed, described, predicted, and explained with models.
Crosscutting Concept	**Patterns** — Students identify the cyclic patterns of lunar phases and the conditions required for eclipses to occur.

CAST Testing Alignment

MULTIPLE CHOICE:

During a lunar eclipse, the Moon appears red instead of disappearing completely. Which best explains this observation?

CONSTRUCTED RESPONSE:

Using a model of the Earth-Sun-Moon system, explain why we see different Moon phases throughout a month and why eclipses don't happen every month.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The Earth-Moon-Sun system creates some of the most dramatic natural phenomena visible from Earth — lunar phases, eclipses, and tides. Understanding these patterns requires modeling how gravity, orbital mechanics, and light interact across vast distances.

Lunar Phases

The Moon completes one orbit around Earth every 29.5 days. As it orbits, the angle between Sun, Moon, and Earth changes, illuminating different portions of the Moon's surface. The Moon doesn't produce light — it reflects sunlight. New moon occurs when the Moon is between Earth and Sun, and full moon when Earth is between Sun and Moon.

Lunar Eclipses

A lunar eclipse occurs when Earth passes directly between the Sun and Moon, casting its shadow on the Moon. The Moon turns red because Earth's atmosphere refracts red wavelengths of sunlight into the shadow. Eclipses don't happen monthly because the Moon's orbit is tilted 5.1 degrees relative to Earth's orbital plane.

Gravity and Orbits

Gravity is the attractive force between all objects with mass. The Moon orbits Earth because gravity provides the centripetal force for orbital motion. The same force creates tides — the Moon's gravity pulls on Earth's oceans, creating bulges on opposite sides of the planet.

Scale and Distance

The Moon is about 384,400 km from Earth. Light from the Sun takes about 8 minutes to reach Earth and an additional 1.3 seconds to reach the Moon. The Moon's diameter is about 1/4 of Earth's, making accurate physical models challenging to build.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Sun angle, Moon position, Earth's shadow, and visible Moon phase as key components of the lunar phase system.

E **Establish Relationships**
Students determine that Moon position and Sun angle together determine Earth's shadow and visible phase through positive and negative relationships.

V **Visualize & Model**
Students build a model showing how the Moon's orbital position creates predictable phase patterns.

E **Evaluate Outcomes**
Students run full moon, eclipse, and new moon scenarios to test their model's predictions against real observations.

R **Revise & Extend**
Students add tidal force or orbital tilt to explore more complex Earth-Moon interactions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Have you ever looked up and seen the Moon turn red? It's not magic — it's science."*

Watch For

Show a time-lapse video of a lunar eclipse if available.

Slide 2: Learning Objectives

3 min

Say: *"Today we're building a model of something that's mystified humans for thousands of years."*

Watch For

Have students read objectives. Pre-teach 'revolution' vs. 'rotation.'

Slide 3: Big Question

3 min

Say: *"Right now, the Moon is out there orbiting us. But WHY does it look different every night?"*

Watch For

Quick poll: Who has seen a blood moon? What did they think was happening?

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model of the Earth-Moon-Sun system to answer this."*

Watch For

Preview each LEVER step. Emphasize that this is a SYSTEM with interacting parts.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key players in this cosmic dance?"*

Watch For

Guide sorting of external vs. internal components.

Slide 6: Activity 2: Connections

8 min

Say: *"When the Moon moves to a certain position, what happens to what we see?"*

Watch For

Help students identify how position determines visibility.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's create a full moon, then an eclipse, then a new moon!"*

Watch For

Students predict first, then run simulations. Compare phases.

Slide 8: Discoveries

5 min

Say: *"So WHY does the Moon turn red instead of disappearing during an eclipse?"*

Watch For

Lead discussion about atmosphere bending light. Connect to sunsets.

Slide 9: STEM Challenge

12 min

Say: *"Now you're building something a planetarium would actually use!"*

Watch For

Groups design Moon phase predictors.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Sun Angle, Moon Position
- **INTERNAL:** Earth's Shadow, Visible Moon Phase

Relationships

- **Sun Angle to Visible Moon Phase:** POSITIVE (+) — When the Sun's angle allows more direct light to hit the Moon's Earth-facing side, we see more of the Moon illuminated (fuller phases).
- **Moon Position to Earth's Shadow:** POSITIVE (+) — When the Moon's position moves into alignment behind Earth, it enters Earth's shadow, creating eclipse conditions.
- **Earth's Shadow to Visible Moon Phase:** NEGATIVE (-) — When Earth's shadow covers the Moon (eclipse), the visible illuminated phase decreases dramatically — the Moon darkens or turns red.

Key Discoveries

- The Moon doesn't produce its own light — it reflects sunlight, so its appearance depends on position
- Lunar eclipses turn the Moon red because Earth's atmosphere bends red light into the shadow
- Eclipses don't happen every month because the Moon's orbit is tilted 5 degrees relative to Earth's
- Gravity is the invisible force that keeps the Moon orbiting Earth and Earth orbiting the Sun

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The Moon makes its own light"	The Moon has no light source — it reflects sunlight. What we call moonlight is actually reflected sunlight. The Moon's brightness depends entirely on its position relative to the Sun.
"Earth's shadow causes Moon phases"	Moon phases are caused by the ANGLE at which we see the Moon's sunlit side, not by Earth's shadow. Earth's shadow only affects the Moon during the rare event of a lunar eclipse.
"The Moon changes shape"	The Moon doesn't physically change — it's always a sphere. What changes is how much of its sunlit side we can see from Earth as it orbits. We're seeing different ANGLES, not different shapes.